Multi-Scale Spectral Kernel Machines for Tabular Data*

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Abstract

We treat supervised learning as the construction of a kernel: once a positive-semidefinite kernel is fixed, kernel ridge regression determines the estimator, so learning is the choice of a data-adapted kernel. We develop the Multi-Scale Spectral Kernel Machine (MS-SKM), a structured, inspectable kernel for tabular data, and study it as a ladder in which each rung lifts one modeling restriction: from an additive model on fixed random Fourier features, to learned frequencies, to a radial kernel on a per-coordinate spectral embedding, to feature mixing, to multi-resolution banks fused on the simplex. Three results organize the design. Interactions come from the kernel nonlinearity, not the linear mixer, which only rotates them from axis-aligned to oblique. The convex multi-bank fuse is positive-semidefinite and decomposes into one component per bank, whereas a linear mixer over banks collapses to a single bank with the pooled frequencies—so the per-bank bandwidth is what fusion adds. A parameter accounting makes the per-band frequency count the overfitting axis and the bank count the multi-scale axis. The ladder isolates each effect on real data, and a variational head trains the same kernel by a sparse Gaussian-process ELBO to return a calibrated predictive distribution.

Keywords: kernel methods, spectral kernels, Gaussian processes, tabular data, kernel ridge regression, additive models, multi-resolution learning, variational inference

1 Introduction

A prediction at a query point borrows strength from training points through a notion of similarity, the *kernel*. Kernel methods fix this object a priori, tree ensembles induce it through a learned partition, and neural networks learn a representation whose inner product is a kernel. Gradient-boosted decision trees remain the default for tabular data and frequently outperform deep models [3, 9, 13, 21]. Taking the kernel itself as the primitive, we ask what a trainable kernel for tabular data should look like, and we build it from the spectral domain as a competitive learned-kernel alternative.

Learnable spectral kernels are not new: spectral-mixture kernels model a spectral density [27], sparse-spectrum Gaussian processes learn finite spectral atoms [15] and à-la-carte kernels learn groups of fast frequencies [29]. Our contribution is an *arrangement* of these decisions into a tabular inductive bias and a *ladder* that makes each decision's contribution measurable. Each rung lifts one restriction:

1. **Rung 0 — fixed-frequency additive model.** Per-coordinate random Fourier features with a linear readout. The frequencies are fixed, the model is additive, there is no kernel. This is a generalized additive model [10] solved in closed form.

*Code available at `https://github.com/asudjianto-xml/Spectral-Kernel`.

2. **Rung A — learned frequencies, still additive.** The frequencies are learned. The model stays additive and kernel-free.
3. **Rung B — the kernel, no mixing.** A radial base kernel on a per-feature spectral embedding, with a block-diagonal encoder so no feature is mixed into another.
4. **Rung 1 — full mixing.** The encoder mixes features. The kernel now sees oblique interactions.
5. **Rung 2 — multi-resolution banks.** Frequencies are partitioned into H banks, each with its own bandwidth, convex-fused on the simplex.

The ladder is the methodological point. A single accuracy number hides whether a model needs interactions at all, whether they are axis-aligned or oblique and whether it needs multiple scales or only more frequencies. Lifting one restriction at a time answers each question separately, on the same data and the same training machinery.

Contributions. We are explicit about what is *proven*, what is a *diagnostic*, and what is a *conjecture supported by a sweep.*

1. *(Proven.)* The per-coordinate feature map realizes a sum of one-dimensional spectral-mixture kernels (Prop. 1), dense in the additive-stationary class between additive Gaussian processes [4] and spectral-mixture kernels [27].
2. *(Proven / diagnostic.)* Interactions come from the radial base kernel, not the mixer (Prop. 2). A linear mixer is a learned metric $W^\top W$ whose off-diagonal blocks are an inspectable diagnostic of *metric* interaction; with the mixer block-diagonal the kernel is still non-additive.
3. *(Proven.)* The convex multi-bank fuse is positive-semidefinite and decomposes exactly into one component per bank in the sum RKHS (Prop. 4), and a single linear mixer over banks collapses to one bank with the pooled frequencies (Prop. 5)—so the per-bank bandwidth, not the bank count, is what convex fusion adds.
4. *(Accounting + sweep.)* The dominant parameter count scales with the per-band frequency count K and is independent of the bank count H (Prop. 6); we observe K to be the overfitting axis and H the multi-scale axis.
5. *(Whittle / diagnostic.)* A one-dimensional criterion for when learning the frequencies helps (Prop. 7): the gradient on a frequency atom is proportional to the local residual spectral mass, vanishing on flat spectra and active on peaked ones.
6. *(Predictive distribution.)* A variational head (Sec. 8) that trains the same learned kernel by the ELBO of a sparse Gaussian process [11, 26], giving a calibrated Gaussian predictive distribution rather than a point.
7. *(Interpretability.)* Because the kernel is learned, its parameters are the explanation: automatic relevance determination (ARD) gives a per-feature importance and the encoder metric $W^\top W$ gives pairwise metric interactions, read directly off the fitted model with no surrogate; a generalized functional ANOVA [12] adds the effect shapes and a variance attribution (Sec. 9).
8. An open-source implementation and an ablation ladder that isolates each effect on real tabular data.

2 The kernel is the object of learning

Let $\{(x_i, y_i)\}_{i=1}^n$ with $x_i \in \mathcal{X} \subseteq \mathbb{R}^d$. A symmetric positive-semidefinite (PSD) kernel k induces a reproducing-kernel Hilbert space (RKHS) $\mathcal{H}_k$ and a Gaussian-process prior $f \sim \mathcal{GP}(0, k)$. Regularized risk minimization in $\mathcal{H}_k$ has, by the representer theorem, the finite solution $\hat{f}(\cdot) = \sum_i \alpha_i k(\cdot, x_i)$; for squared loss this is kernel ridge regression (KRR), which coincides with the GP posterior mean

$$\hat{f}(x) = k(x, X)\,(K + \lambda I)^{-1} y, \qquad K_{ij} = k(x_i, x_j). \tag{1}$$

The estimator (1) is a fixed functional of k; the learning problem is the choice of k from a parameterized family $\{k_\theta\}$, here by empirical Bayes over θ [23]. We therefore study the admissible k_θ directly.

3 A per-coordinate spectral construction

A continuous stationary kernel $k(x, x') = \kappa(x - x')$ is PSD if and only if κ is the Fourier transform of a finite nonnegative measure μ, the spectral measure (Bochner). Kernel learning restricted to the stationary class is spectral-measure estimation [27]. Rather than estimate μ on $\mathbb{R}^d$ jointly, MS-SKM estimates a measure on each coordinate. Fix per-coordinate frequencies $\{\omega_{j,k}\}_{k=1}^K$, nonnegative amplitudes $\{a_{j,k}\}$ and a per-coordinate automatic-relevance-determination (ARD) scale $s_j > 0$ [17, 19, 23], and define the feature map with components

$$\psi(x)_{j,k} = a_{j,k}\big[\cos(2\pi s_j\,\omega_{j,k}\,x_j),\ \sin(2\pi s_j\,\omega_{j,k}\,x_j)\big], \qquad j = 1, \dots, d,\ k = 1, \dots, K. \tag{2}$$

Unlike random Fourier features [22], where each feature is $\cos(\omega^\top x)$ with $\omega \in \mathbb{R}^d$ mixing all coordinates, (2) is a one-dimensional Fourier expansion applied independently to each coordinate—the periodic numerical embedding of tabular deep learning [8, 30], read spectrally.

Proposition 1 (Additive spectral-mixture realization)**.** *The inner product of* (2) *is the additive kernel*

$$\langle \psi(x), \psi(x') \rangle = \sum_{j=1}^{d} \sum_{k=1}^{K} a_{j,k}^2\, \cos\big(2\pi s_j \omega_{j,k}(x_j - x_j')\big) = \sum_{j=1}^{d} \kappa_j(x_j - x_j'), \tag{3}$$

where each κ_j is a one-dimensional stationary kernel with atomic spectral measure

$$\mu_j = \tfrac{1}{2} \sum_k a_{j,k}^2 \big(\delta_{s_j \omega_{j,k}} + \delta_{-s_j \omega_{j,k}}\big).$$

Hence ψ realizes a sum of per-coordinate spectral-mixture kernels and is PSD. As $K \to \infty$ with $\{a_{j,k}^2\}$ a consistent quadrature of a density p_j, each κ_j converges to the kernel of p_j, so the family is dense in the additive-stationary class [4, 27].

Proof. The identity $\cos A \cos B + \sin A \sin B = \cos(A - B)$ with $A = 2\pi s_j \omega_{j,k} x_j$ and $B = 2\pi s_j \omega_{j,k} x_j'$ gives the cosine term after the $a_{j,k}^2$ weighting; summing over k then j yields (3). Each μ_j is finite, nonnegative and symmetric, so by Bochner κ_j is PSD; a sum of PSD kernels in disjoint coordinates is PSD. Density is coordinatewise [27] and additive [4]. □

With a linear readout $f(x) = \beta^\top \psi(x)$ this is rung 0 (frequencies fixed, solved in closed form) and rung A (frequencies learned by gradient descent). In both, the readout weights are the per-frequency amplitudes, the model is additive, and there is no kernel.

4 Interactions from the kernel, not the mixer

The additive map (3) has no cross-coordinate interactions. We introduce a single linear encoder W, $\phi(x) = W\psi(x)$, and a Laplace base kernel,

$$k_\theta(x, x') = \exp\big(-\|W(\psi(x) - \psi(x'))\|_2/T\big), \qquad \theta = (s, \omega, a, W, T), \tag{4}$$

the deep-kernel construction [28] with a shallow, linear warp on additive spectral features. It is essential to separate three notions of interaction. *Metric:* cross-coordinate blocks of $M = W^\top W$. *Kernel:* failure of k_θ to be additive. *Functional:* a non-additive contribution to the fitted predictor.

Proposition 2 (Where interactions come from)**.** *Let $\Delta\psi = \psi(x) - \psi(x')$, blocked by coordinate. Then k_θ in* (4) *is PSD and*

$$\|W\Delta\psi\|_2^2 = \Delta\psi^\top M \Delta\psi = \sum_j \Delta\psi^{(j)\top} M_{jj} \Delta\psi^{(j)} + \sum_{j \neq j'} \Delta\psi^{(j)\top} M_{jj'} \Delta\psi^{(j')}, \quad M = W^\top W \succeq 0. \tag{5}$$

Consequently (i) *a linear readout on ψ (no kernel) is additive for any W, since $\beta^\top W\psi$ is linear in ψ and each block of ψ depends on one coordinate, so a linear mixer neither creates nor removes interactions;* (ii) *with M block-diagonal the kernel is still non-additive, because* $\exp(-\sqrt{\sum_j d_j^2}/T)$ *does not factor over the d_j, so kernel and functional interactions are present with zero metric interaction;* (iii) *the off-diagonal blocks $M_{jj'}$ are an inspectable diagnostic of* metric *interaction, and mixing rotates the interactions the kernel forms from axis-aligned to oblique.*

Proof. $\exp(-\|Lz\|/T)$ is PSD for any matrix L and $T > 0$: $\exp(-\|u\|/T)$ is PSD (Matérn-$\frac{1}{2}$) and composition with a linear map preserves positive-definiteness; take $L = W$. The decomposition is the block expansion of $\Delta\psi^\top W^\top W \Delta\psi$. Claim (i) is immediate from linearity. For (ii), with M block-diagonal the squared distance is $\sum_j d_j^2$ but the Laplace map of its square root does not separate. $\square$

Proposition 3 (Universality)**.** *If the per-coordinate bank separates points and $\phi = W\psi$ is injective on the compact domain, then k_θ is universal: its RKHS is dense in $C(\mathcal{X})$ [18]. Every continuous function, and thus every interaction, is approximable; the additive base is not a structural barrier. Interactions enter through the radial nonlinearity acting on linear projections of the univariate features, exactly as an RBF kernel on raw inputs realizes interactions despite a coordinatewise input.*

Rung B is (4) with W block-diagonal (a per-feature encoder, no mixing); rung 1 is (4) with W full. By Prop. 2 rung B already has interactions, axis-aligned; rung 1 makes them oblique.

5 Multi-resolution fusion

We replicate the construction in H banks. Bank h carries its own frequencies ω_h and amplitudes a_h initialized on a band of a log-spaced partition of the frequency range, its own bandwidth T_h, and a shared encoder W applied to each bank's own $2dK$-dimensional feature vector $\psi_h(x)$ separately. The banks are fused convexly,

$$k(x, x') = \sum_{h=1}^{H} w_h\, k_h(x, x'), \qquad w = \mathrm{softmax}(\cdot) \in \Delta^{H-1}, \tag{6}$$

with k_h as in (4) using (ω_h, a_h, T_h). This is rung 2.

Proposition 4 (Closure and decomposition)**.** *For $w_h \geq 0$, k in (6) is PSD. The fused KRR fit decomposes exactly into one component per bank,*

$$\hat{g}_h = w_h K_h (K + \lambda I)^{-1} y, \qquad \textstyle\sum_h \hat{g}_h = \hat{f}, \tag{7}$$

and, by Aronszajn's sum-of-kernels theorem [1, 2], the RKHS of $k = \sum_h w_h k_h$ is the sum space $\sum_h \mathcal{H}_{k_h}$ with the infimal-convolution norm $\|f\|^2_{\mathcal{H}_k} = \min_{f=\sum_h g_h} \sum_h \frac{1}{w_h} \|g_h\|^2_{\mathcal{H}_{k_h}}$. If each bank is normalized ($k_h(x,x) = 1$) the posterior mean is invariant along $(w, \lambda) \mapsto (\alpha w, \alpha \lambda)$, so the simplex fixes the mixture scale.

Proof. A nonnegative combination of PSD matrices is PSD. Equation (7) is Gaussian conditioning: with independent priors $g_h \sim \mathcal{GP}(0, w_h k_h)$ and $\mathrm{Cov}(g_h, y) = w_h K_h$, $\mathbb{E}[g_h \mid y] = w_h K_h (K + \lambda I)^{-1} y$, and $\sum_h w_h K_h = K$. The sum-RKHS norm is Aronszajn's theorem. Invariance follows from $\alpha K(\alpha K + \alpha \lambda I)^{-1} = K(K + \lambda I)^{-1}$. □

The decomposition is the rigorous statement that a convex fuse of banks at different scales is *not* reducible to a single kernel. The next proposition shows what the alternative — mixing the banks linearly — actually is.

Proposition 5 (A linear mixer over banks collapses to one bank)**.** *Let the banks be combined by a single linear map applied to the concatenation $[\psi_1(x); \ldots; \psi_H(x)]$, followed by one base kernel of bandwidth T. The concatenation of H banks of K frequencies is a single bank of HK frequencies, and one linear encoder over it is the encoder of (4) with $K' = HK$. Hence "H banks + one linear mixer" equals "one bank with $K' = HK$ frequencies": a single kernel with a single bandwidth. The per-bank bandwidths T_h of (6) are absent from this construction, and a single bandwidth cannot span frequencies of widely separated scale.*

Proposition 6 (Parameter accounting)**.** *With H banks of K frequencies each and the shared per-bank encoder $W \in \mathbb{R}^{d_\phi \times 2dK}$, the free-parameter count is*

$$\underbrace{H \cdot d \cdot 2K}_{\textit{bank spectra}} + \underbrace{d \cdot 2K \cdot d_\phi}_{\textit{shared encoder } W} + O(H), \tag{8}$$

so the dominant encoder term scales with K and is independent of H.

Remark 1 (Banks versus frequencies)**.** The two axes generalize differently. Increasing H adds multi-resolution structure on a shared basis, gated by the simplex weights w_h. Increasing K widens the dominant ungated encoder (8) and packs more frequencies into a band whose amplitude envelope is already a smooth quadrature (Prop. 1). We therefore expect K to be the overfitting axis and H the multi-scale axis, a multi-resolution echo of the frequency-overfitting of [6]. The selection of w here is by marginal likelihood, jointly with the banks; selecting w by an unbiased risk estimate over fixed banks [7, 25] would inherit the oracle guarantee that convex fusion is never worse than the best single bank, and is a natural alternative.

6 When does learning the frequencies help?

Sparse-spectrum methods learn ω by marginal likelihood and overfit on smooth targets [6, 15]. We give a partial criterion in the cleanest setting, one-dimensional regression, and use it as a tabular diagnostic.

Proposition 7 (Frequency-learning gradient, 1-D Whittle regime)**.** *Under the Whittle approximation to the Gaussian log-likelihood, with model spectrum $S_\theta(\omega) = \lambda + \sum_k a_k^2 g(\omega - \omega_k)$, the gradient moving an atom is*

$$\frac{\partial \mathcal{L}}{\partial \omega_k} \;\propto\; a_k^2\, \partial_\omega r(\omega)\big|_{\omega_k}, \qquad r(\omega) = \hat{p}(\omega) - \textstyle\sum_{k'} a_{k'}^2 g(\omega - \omega_{k'}), \tag{9}$$

the residual spectral density ($\hat{p}$ the periodogram, g a fixed atom kernel). If $\hat{p}$ is locally flat the atoms do not move and a fixed grid is already a consistent quadrature; if $\hat{p}$ is peaked the atoms migrate to the lines and a learned grid strictly improves the fit.

Sketch. Differentiate $-\frac{1}{2}\int[\log S_\theta + \hat{p}/S_\theta]d\omega$ in ω_k using $\partial S_\theta/\partial\omega_k = -a_k^2 g'(\omega - \omega_k)$; the leading term is $a_k^2 \partial_\omega r(\omega_k)$. □

For multivariate tabular data the per-coordinate spectral density is not canonically defined, so we read this as a diagnostic: learning the frequencies helps in proportion to per-coordinate spectral peakedness. This is consistent with the smooth-versus-periodic reversal in Section 9.

7 Training and inference

Hyperparameters θ are fit by the negative log marginal likelihood $\mathcal{L} = \frac{1}{2} y^\top (K + \lambda I)^{-1} y + \frac{1}{2}\log|K + \lambda I| + \text{const}$, optimized by gradient descent on random support subsets (Cholesky in double precision, autograd); the noise λ is the learned ridge. Prediction is full-train KRR (1), solved by a truncated Lanczos tridiagonalization [14] so the cost is a handful of matrix-vector products rather than a dense factorization. The frequencies and amplitudes are learned, never frozen, so the construction is not random Fourier features: we learn the kernel rather than approximate a fixed one. The feature and label scalers are fit on train, the ridge and the calibration temperature on a validation fold, and test data is touched only at prediction.

8 Predictive distributions via a variational bound

The NLML objective with the KRR decode returns the Gaussian-process posterior mean — a point prediction. The exact GP posterior also has a closed-form predictive variance, but it requires a dense solve and is tied to the Gaussian likelihood. To obtain a predictive distribution that scales by minibatching and extends to non-Gaussian likelihoods, we train the same learned kernel by the evidence lower bound (ELBO) of a sparse variational Gaussian process [11, 26].

Place M inducing inputs Z with prior $p(u) = \mathcal{N}(0, K_{ZZ})$, and a whitened variational posterior $q(u) = \mathcal{N}(Lm,\, LSL^\top)$ with $L = \text{chol}(K_{ZZ})$. The variational marginal at a point is $q(f_i) = \mathcal{N}(\mu_i, s_i^2)$ with

$$\mu_i = b_i^\top m, \qquad s_i^2 = k_{ii} - b_i^\top (I - S)\, b_i, \qquad b_i = L^{-1} k(Z, x_i), \tag{10}$$

and the kernel's unit diagonal gives $k_{ii} = 1$. For a Gaussian likelihood with noise σ^2 the bound is

$$\text{ELBO} = \sum_i \Big[\log \mathcal{N}(y_i \mid \mu_i, \sigma^2) - \tfrac{s_i^2}{2\sigma^2}\Big] - \text{KL}\big(q(u)\,\|\,p(u)\big), \quad \text{KL} = \tfrac{1}{2}\big[\,\text{tr}\, S + \|m\|^2 - M - \log|S|\big]. \tag{11}$$

The spectral parameters, the inducing inputs Z and the variational parameters (m, S, σ^2) are learned jointly by maximizing the ELBO on minibatches. The predictive distribution at $x^\star$ is Gaussian, $\mathcal{N}(\mu^\star,\, s^{\star 2} + \sigma^2)$, so every prediction carries a variance.

Table 1: `make_moons` test metrics. AUC and F1 higher is better; Brier and log loss lower is better. AUC saturates; the kernel rungs lead on F1 and Brier, and CatBoost is the least calibrated (highest Brier and log loss) despite a comparable AUC.

model	AUC	F1	Brier	LogLoss
rung 0: SpectralGAM	0.9895	0.9385	0.0430	0.1386
rung A: LearnedGAM	0.9907	0.9457	0.0430	0.1537
rung B: MSSKM(mix=False)	0.9888	0.9605	0.0358	0.1444
rung 1: MSSKM()	0.9889	0.9605	0.0350	0.1393
rung 2: MSSKM(H=4)	0.9896	0.9605	0.0367	0.1423
CatBoost	0.9893	0.9444	0.0467	0.2011

The intervals are calibrated. On California Housing the 90% and 95% central predictive intervals cover 0.93 and 0.95 of held-out points; on a one-dimensional series with a held-out gap the predictive standard deviation grows from in-sample through the gap to extrapolation, the expected Gaussian-process behavior. The sparse M-point head reaches $R^2 \approx 0.84$ where the exact decode reaches 0.88. The same bound extends to a Bernoulli or softmax likelihood for classification.

9 Experiments

We use California Housing (regression, n=20,640, d=8) and Breast Cancer (binary, n=569, d=30) from scikit-learn, and an hourly bike-sharing series (regression, periodic), each with a held-out 20% test split; the remaining data is split into train and validation for early stopping and the ridge. Classification uses one-hot KRR with a temperature-scaled posterior. The kernel is Laplace, $K = 16$ frequencies per feature per bank unless swept, $d_\phi = 128$.

An illustrative 2-D example. Figure 1 fits the ladder and a tuned CatBoost on `make_moons`, two interleaving crescents whose separation needs an oblique interaction. The additive rungs (0, A) draw blocky, near-separable boundaries; the boundary curves the moment the kernel appears (rung B), feature mixing (rung 1) wraps it obliquely around both crescents and CatBoost draws an axis-aligned staircase. Accuracy sits at the noise floor, so the metrics (Table 1) read through the boundary: AUC saturates, the kernel rungs win on F1 and Brier, and CatBoost — comparable on AUC — has markedly worse Brier and log loss, the signature of an accurate ranker with overconfident probabilities. The variational head (Sec. 8) turns the same kernel into a predictive distribution (Figure 2): the predictive standard deviation is low over the data-covered crescents and rises in the empty regions, the epistemic uncertainty of a Gaussian process.

The ladder decomposition. Table 2 reports each rung. On California the gain over the additive model is almost entirely the kernel and almost entirely axis-aligned: rung B (kernel, no mixing) recovers $0.711 \to 0.873$ of the $0.711 \to 0.876$ total, and the oblique mixing of rung 1 adds 0.003. On Breast Cancer the reverse holds: the kernel with no mixing adds nothing ($0.947 \to 0.947$) and the whole gain is the oblique mixing ($\to 0.965$). The interactions California needs are axis-aligned and reached by the radial kernel without mixing (Prop. 2); those Breast Cancer needs live along learned feature combinations.

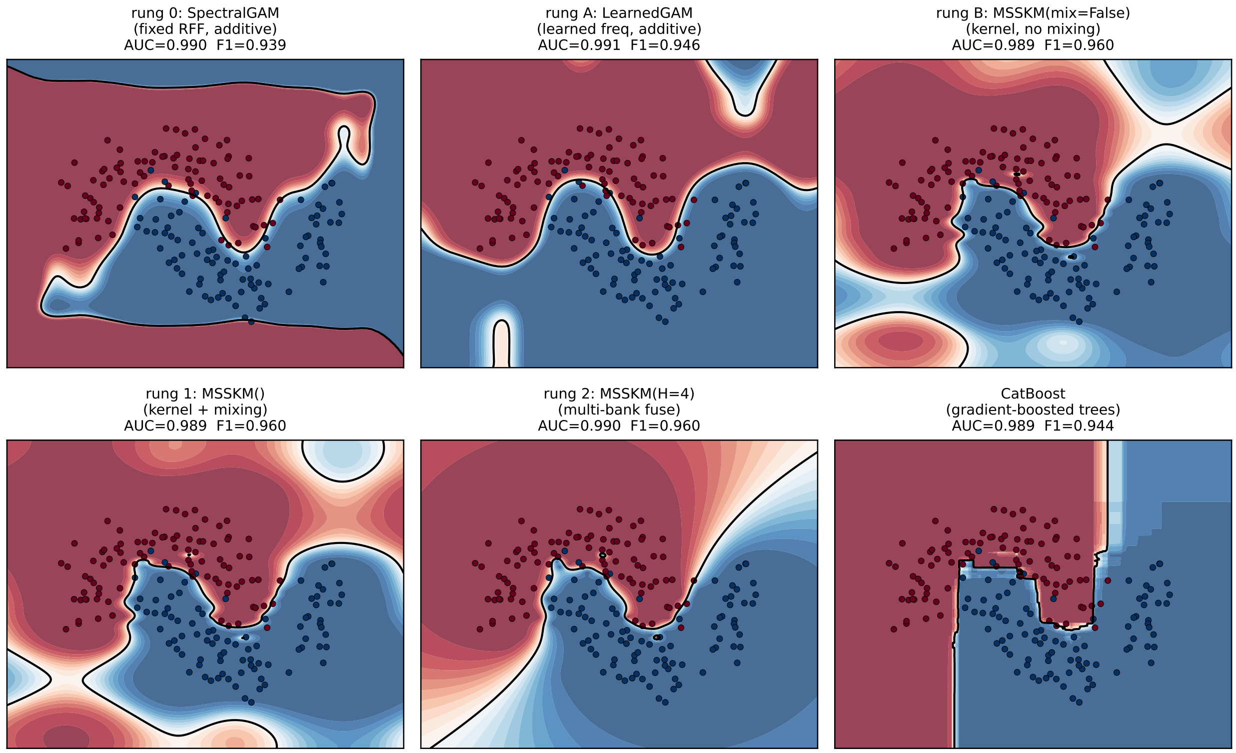

Figure 1: Decision boundaries on `make_moons` (shading is $P(\text{class } 1)$, black line the 0.5 boundary, dots the test set). Additive rungs give blocky separable boundaries; the kernel curves them (rung B), mixing makes them oblique (rung 1); CatBoost is an axis-aligned staircase.

Learning the frequencies (rung 0 vs A). On smooth California, learning the frequencies moves test R^2 by < 0.01 ($0.713 \to 0.711$), and on Breast Cancer it is unchanged. This is the flat-spectrum case of Prop. 7: a fixed grid is already a consistent quadrature and the amplitudes carry the information.

Convex fuse versus a linear mixer (Prop. 5). At a matched frequency budget of 256 per feature, one bank with $K{=}256$ (a linear mixer over the pooled frequencies) matches one bank with $K{=}64$ on California (0.876 vs 0.876) and is worse on Breast Cancer (0.939 vs 0.965, overfitting), while splitting the same 256 frequencies into four convex-fused banks of $K{=}64$ reaches 0.885 on California. Pooling frequencies into one kernel does nothing; the gain is the per-bank bandwidth of the convex fuse, exactly Prop. 5.

The $H \times K$ sweep. Table 3 sweeps banks against per-band frequencies, and bears out Prop. 6 and Remark 1. Along each row (fixed H, increasing K) the metric is flat or declines: extra frequencies in one bank do not help and the ungated encoder eventually overfits, sharpest on small Breast Cancer where $K{=}32$ at $H{=}8$ falls back. Down each column (fixed K, increasing H) the multi-scale regressions improve monotonically (California $0.873 \to 0.885$, Bike-hourly $0.943 \to 0.954$ at $K{=}8$), while small Breast Cancer is flat-to-worse, its near-uniform fusion weights signalling no multi-scale structure to exploit. The best corner is moderate K with larger H: capacity comes from banks, not from frequencies per bank.

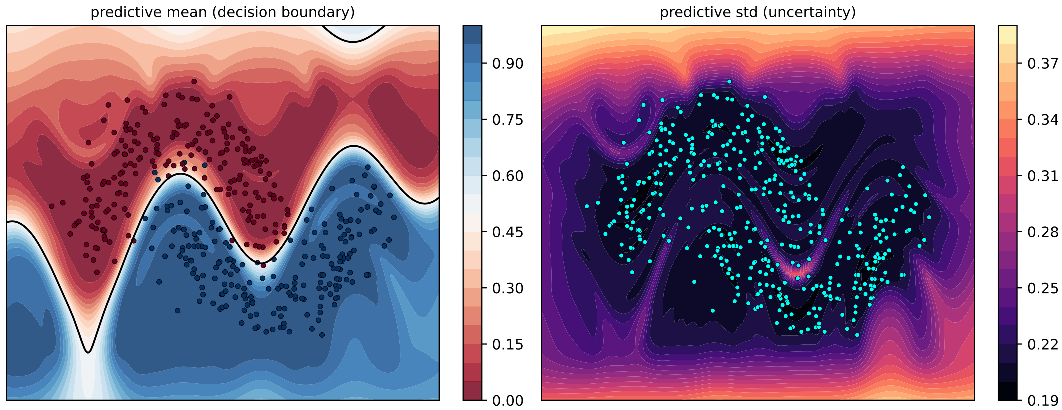

Figure 2: The variational (ELBO) head on `make_moons`, regressing the 0/1 labels. Left: predictive mean with the 0.5 boundary. Right: predictive standard deviation — low over the two crescents where there is data, high in the empty regions. This is the latent predictive spread, not a calibrated Bernoulli posterior; a classification likelihood is future work.

Table 2: The ladder, test metric (R^2 for California, accuracy for Breast Cancer). Each rung lifts one restriction.

rung	model	California R^2	Breast Cancer acc
0	fixed RFF, additive, no kernel	0.713	0.947
A	learned freq, additive, no kernel	0.711	0.947
B	kernel, no mixing (block-diagonal W)	0.873	0.947
1	kernel, full mixing	0.876	0.965
2	multi-bank convex fuse (H=4)	0.885	0.956

Interpretability from the learned kernel. The kernel is learned, so its parameters *are* the explanation: the quantities below are closed-form functionals of the fitted parameters, computed exactly, with no surrogate model, perturbation, sampling or permutation (Figure 3). *Relevance.* The scale s_j in (2) is the inverse length scale of feature j: it multiplies x_j inside every $\cos / \sin(2\pi s_j \omega_{j,k} x_j)$, so as $s_j \to 0$ those coordinates flatten to constants and feature j leaves the kernel. Training therefore drives s_j small for uninformative features and large for relevant ones, which is automatic relevance determination [17, 19]. *Importance.* Relevance alone ignores how much spectral energy a feature carries, so we score feature j by how strongly the embedding turns with it. Differentiating (2), $\partial_{x_j} \psi(x)_{j,k} = 2\pi s_j \omega_{j,k} a_{j,k} \left[-\sin, \cos\right](2\pi s_j \omega_{j,k} x_j)$, whose squared norm is *exactly* $(2\pi)^2 s_j^2 \omega_{j,k}^2 a_{j,k}^2$ — the cosine and sine of one frequency share a constant gradient norm ($\sin^2 + \cos^2 = 1$), independent of x, so no average is taken. Summing over the K frequencies and the convex bank weights w_h,

$$I_j \;=\; (2\pi)^2 \, s_j^2 \sum_h w_h \sum_k a_{h,j,k}^2 \, \omega_{h,j,k}^2 \;=\; \int (2\pi\xi)^2 \, d\mu_j(\xi), \tag{12}$$

the second moment of feature j's learned spectral measure μ_j (Prop. 1), equal to the prior mean-square gradient $\mathbb{E}[(\partial_{x_j} f_j)^2]$ of its additive component. It is exact in closed form and comparable across features because the inputs are standardized; the marginal variance $\kappa_j(0) = \sum_k a_{j,k}^2$ is exact in the same way. (We report I_j normalized to sum to one.) On California the importance concentrates on the geographic coordinates, whose effect on price is sharp and local; a strong but

Table 3: Test metric over H (banks) $\times$ K (frequencies per feature): R^2 for California and Bike-hourly, accuracy for Breast Cancer. **Bold** is the configuration selected on the validation fold and evaluated once on test. Generated by `benchmarks/sweep_hk.py`.

Dataset	H	K=8	K=16	K=32
California (R^2)	1	0.873	0.875	0.875
	2	0.881	0.880	0.882
	4	**0.883**	0.883	0.882
	8	0.885	0.884	0.884
Breast Cancer (acc)	1	0.956	0.974	0.956
	2	0.965	0.965	**0.982**
	4	0.956	0.956	0.965
	8	0.947	0.974	0.965
Bike-hourly (R^2)	1	0.943	0.942	0.944
	2	0.951	0.952	0.946
	4	0.953	0.953	0.953
	8	0.954	**0.956**	0.953

smooth feature such as median income carries high relevance yet low gradient energy. The off-diagonal blocks of the encoder metric $M = W^\top W$ (Prop. 2) expose which pairs the model couples — here the household-structure features (average occupancy with average bedrooms and rooms) and the latitude–longitude pair. This is the metric interaction of Prop. 2 made visible; with W block-diagonal (rung B) it is identically zero.

Functional ANOVA of the fitted predictor. Relevance and importance are closed-form functionals of the kernel; to read the *shape* of each effect and attribute prediction variance we decompose the fitted predictor itself, under the empirical joint measure of the training set, into a grand mean, per-feature main effects and pure pairwise interactions, $f(x) = f_0 + \sum_j g_j(x_j) + \sum_{j<k} g_{jk}(x_j, x_k) + r(x)$ — the generalized functional ANOVA for dependent inputs [12]. Because the kernel does not factor over coordinates this decomposition is numerical rather than closed form: the components are fit to the model's own predictions by penalized backfitting over per-feature quantile bins, and each interaction is *purified* so that $\mathbb{E}[g_{jk} \mid x_j] = \mathbb{E}[g_{jk} \mid x_k] = 0$, making it orthogonal to its own main effects [16]. The fit uses only feature combinations that occur in the data, so it never queries the model off the data manifold — the failure mode of the marginal (interventional) partial-dependence reading [5] under correlated features. The variance attribution then balances exactly,

$$\mathrm{Var}(f) \;=\; \sum_j \mathrm{Var}(g_j) \;+\; \sum_{j<k} \mathrm{Var}(g_{jk}) \;+\; \mathrm{cov} \;+\; \mathrm{rem}, \tag{13}$$

where cov is the between-component dependence (zero when the features are independent) and rem the order-three-and-higher and within-bin variation; both are reported rather than absorbed. On California (Figure 4) the additive main effects of median income and the two geographic coordinates carry the bulk of the explained variance, the pairwise interactions are an order of magnitude smaller (the largest couple average occupancy to location), the strong geographic collinearity surfaces as a large negative covariance (-0.42 of the variance), and a fifth of the variance is genuinely higher order. This complements the closed-form ARD reading: importance ranks features by the kernel's

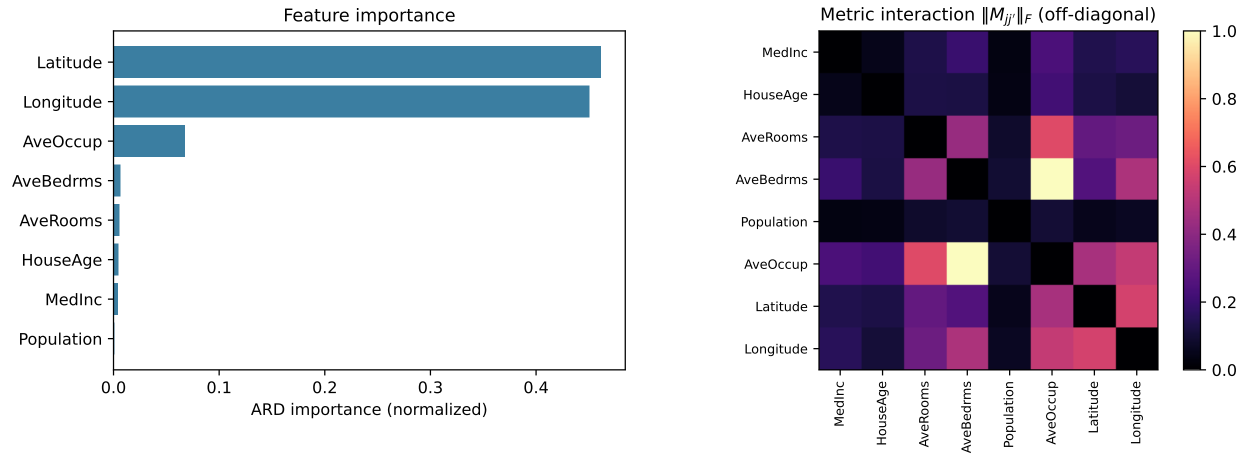

Figure 3: ARD interpretability on California Housing, read directly from the fitted kernel (H=4, K=8). Left: per-feature importance I_j of (12). Right: metric interaction $\|M_{jj'}\|_F$ between feature pairs, diagonal removed.

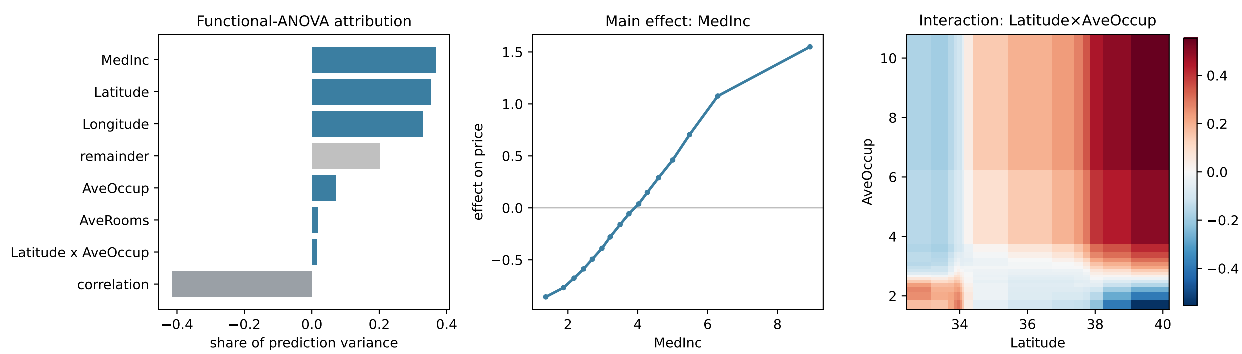

Figure 4: Functional-ANOVA decomposition of the fitted MS-SKM on California Housing (H=4, K=8), under the empirical training measure. Left: variance attribution (13) — main effects, pairwise interactions, the between-component covariance and the higher-order remainder. Center: the purified main effect of median income. Right: the strongest pairwise interaction surface.

mean-square gradient, while the functional ANOVA shows the form those effects take and how feature dependence redistributes them.

10 Related work

Spectral-mixture [27], sparse-spectrum [15] and à-la-carte [29] kernels learn d-dimensional frequencies that couple all coordinates; (2) is deliberately the opposite, additive before any mixing, with interactions introduced by an inspectable operator. Non-stationary spectral kernels [24] relax stationarity. Deep kernel learning [28] warps inputs by a deep network and can collapse [20]; our warp is shallow and linear by design, which restricts and exposes the collapses rather than precluding them. The per-coordinate Fourier embedding is the periodic numerical embedding of tabular deep learning [8, 30], here given a spectral reading. Convex kernel fusion has the exact decomposition and oracle guarantees of the sum RKHS [1, 2, 25]. Gradient-boosted trees remain the tabular default [3, 9, 13, 21].

11 Discussion

The ladder turns one model into a sequence of controlled questions. Interactions come from the kernel, not the mixer; the mixer only rotates them. Multiple banks beat more frequencies only because the convex fuse keeps a bandwidth per scale; a linear mixer over banks is one wider bank. The per-band frequency count is the overfitting axis and the bank count the multi-scale axis. These are not asymptotic claims but statements read off real data by lifting one restriction at a time. Three directions remain: a broader multi-seed benchmark against gradient boosting, fusion weights chosen by an unbiased risk estimate to inherit the oracle guarantee and a low-rank input projection before the per-coordinate map for genuinely multivariate frequencies.

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